

Figure 1

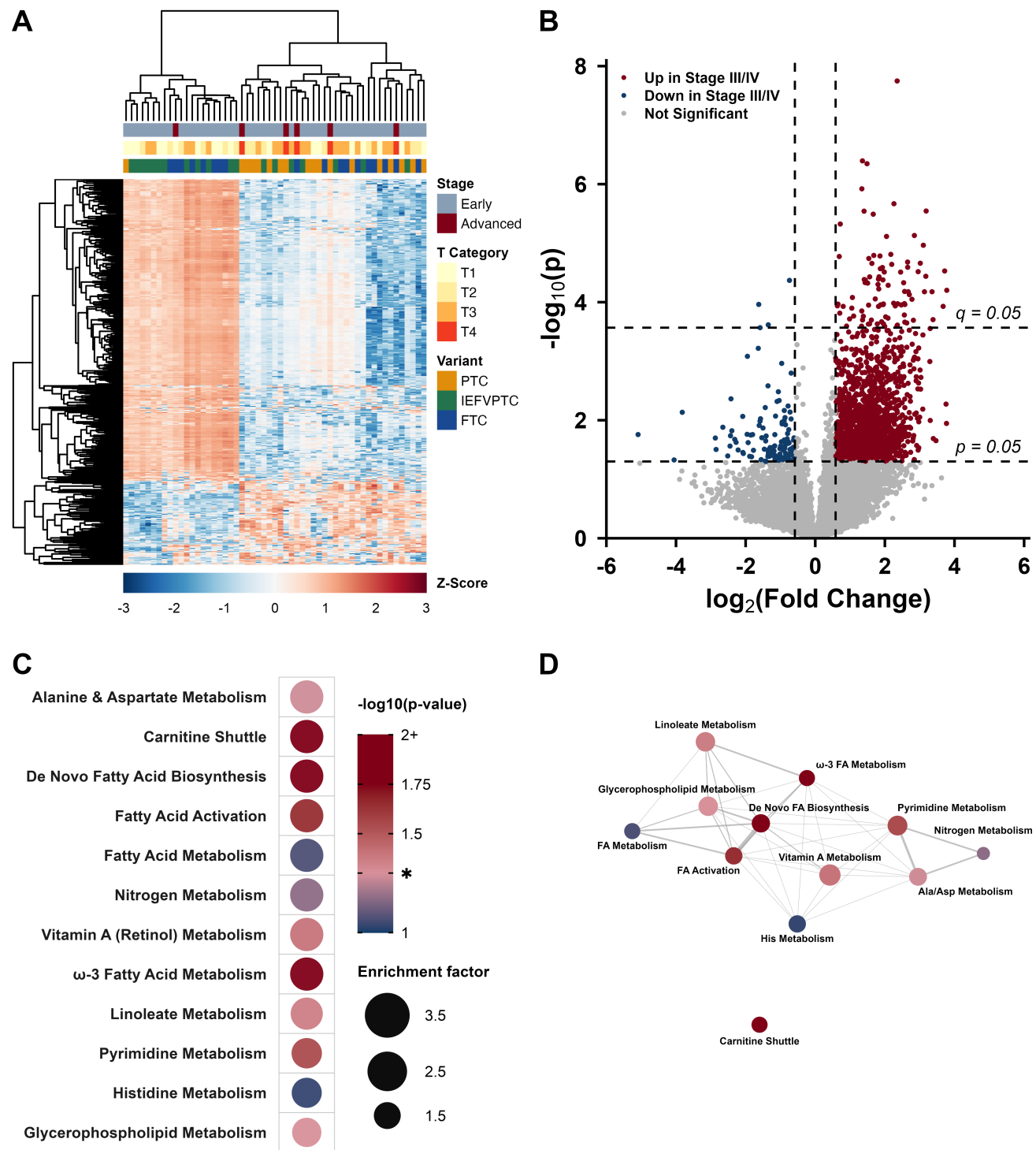
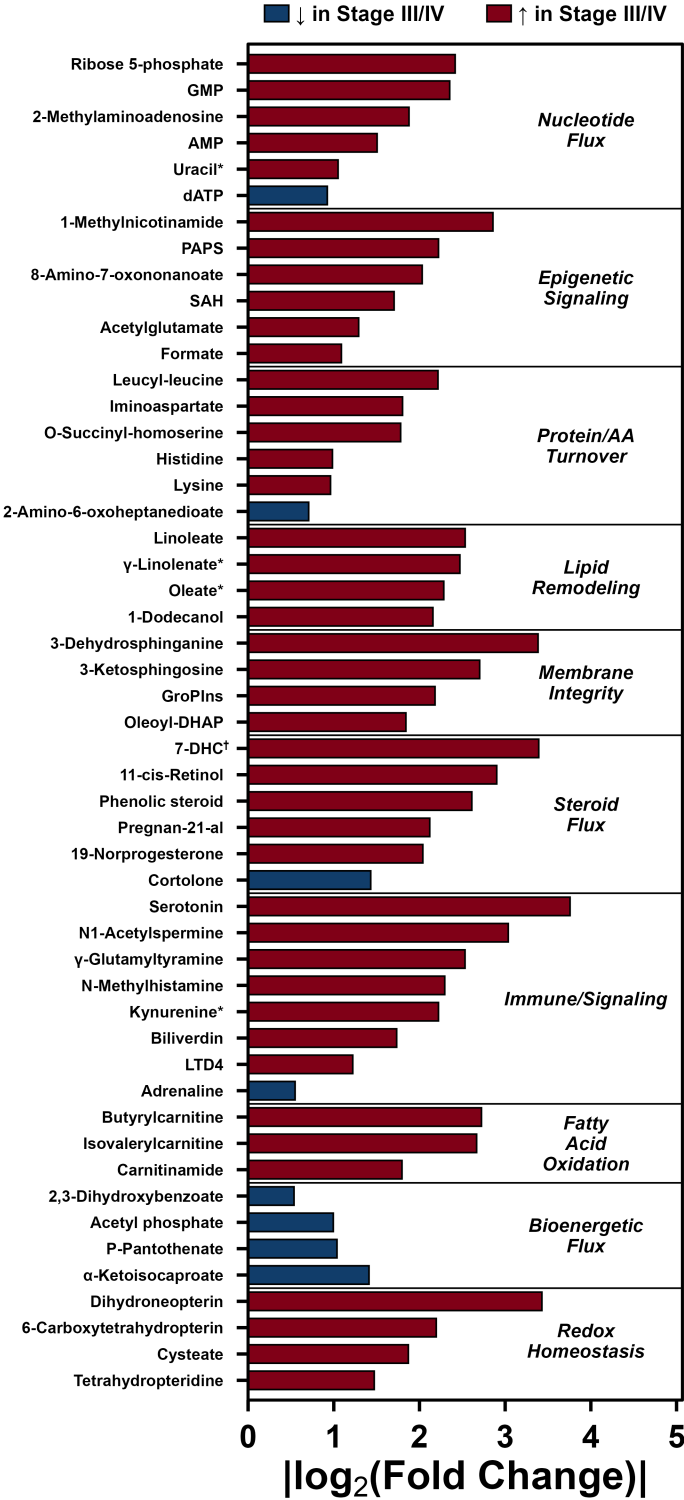
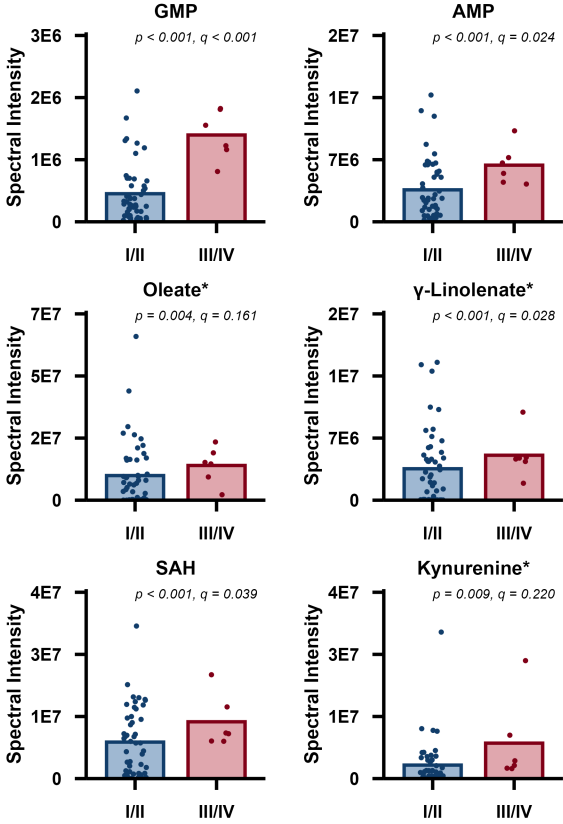


Figure 2

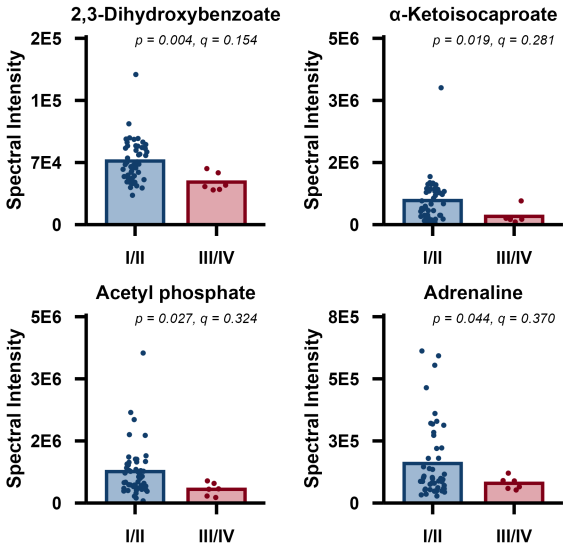
A



B



C



A

Heatmap showing Spearman Correlation between metabolite levels. The color scale ranges from -1 (blue) to 1 (red). The metabolites listed on the axes are: 11-cis-Ret, Oleate*, γ-Linolenate*, 7-DHC, Ribose 5-phosphate, SAH, Histidine, Lysine, Butyrylcarnitine, Kynurenine*, Citrulline, LTD4, Serotonin, Acetylglutamate, AMP, GMP, Adrenaline, Acetyl phosphate, 2,3-Dihydroxybenzoate, and α-Ketoisocaproate.

B

Scatter plots showing correlations between metabolite levels:

- Acetyl phosphate vs GMP: $\rho = -0.45$, $p < 0.001$
- Adrenaline vs 3-Ketosphingosine: $\rho = -0.72$, $p < 0.001$
- Adrenaline vs Kynurenine*: $\rho = -0.42$, $p = 0.001$
- Adrenaline vs SAH: $\rho = -0.52$, $p < 0.001$

C

Scatter plots showing correlations between metabolite levels:

- GMP vs Ribose 5-phosphate: $\rho = 0.54$, $p < 0.001$
- 3-Ketosphingosine vs Oleoyl-DHAP: $\rho = 0.67$, $p < 0.001$
- 1-Methylnicotinamide vs SAH: $\rho = 0.67$, $p < 0.001$
- PAPS vs SAH: $\rho = 0.61$, $p < 0.001$
- Acetylglutamate vs Citrulline: $\rho = 0.53$, $p < 0.001$
- Kynurenine* vs Serotonin: $\rho = 0.39$, $p = 0.003$

D

Scatter plots showing correlations between metabolite levels:

- SAM/SAH ratio vs 1-Methylnicotinamide: $\rho = -0.44$, $p < 0.001$
- SAM/SAH ratio vs γ-Linolenate*: $\rho = -0.33$, $p = 0.013$
- SAM/SAH ratio vs GMP: $\rho = -0.24$, $p = 0.073$
- SAM/SAH ratio vs AMP: $\rho = -0.24$, $p = 0.083$

Figure 4

